

Inside CodeWarrior

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Goal: learn tips and techniques to progress beyond the basics with Metrowerks CodeWarrior for Palm handhelds.



Agenda

- Project Stationery
- Build Targets and Prefix Files
- PalmRez
- The Editor: More than meets the eye
- Debugging
- Resources: Rez and PilRC

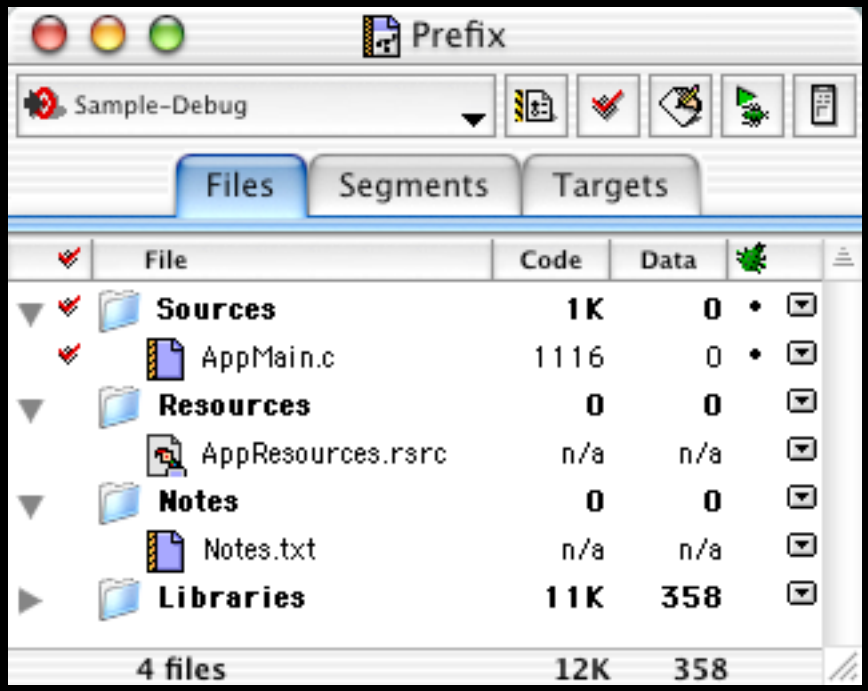


Project Stationery

- Personalized for each developer
- Elements of the perfect stationery project:
  - Release and debug targets
  - Testing target (Host Control)
  - Constructor templates
  - Utility static libraries
  - Engineering notes
  - Basic features (About, menus, prefs)
- Work in progress

Build Targets and Prefix Files

- Build targets
  - Common source code
  - Different settings
  - Debug, release, POSE
- Prefix files
  - Common header files
  - Compiler flags
  - Pragmas
- C/C++ Language target panel



Example Prefix File

```
// Prefix file for debug target

#ifndef PILOT_PRECOMPILED_HEADERS_OFF
    #define PILOT_PRECOMPILED_HEADERS_OFF
#endif

#include <PalmOS.h>                 // Must be first.

// Define compiler variables to control conditional compilation
#define CHECK_ASSERTIONS            1
#define LOG_DB                      0
#define FILL_DM_MEMORY              0
#define PREREGISTERED               1
#define REINIT_USER_PREFERENCES     0
#define REINIT_REGISTRATION         0

// Compiler can assume vars not changed by pointer aliases
#pragma side_effects off

// Include common project files
#include "Assert.h"
```



Build Targets

- Create Debugging and Release targets
  - Build the debug target
  - Clone to build release target
- Panels
  - 68K Processor
  - Global Optimizations
  - PalmRez Post Linker
  - C/C++ Language



Target Setting—68K Processor

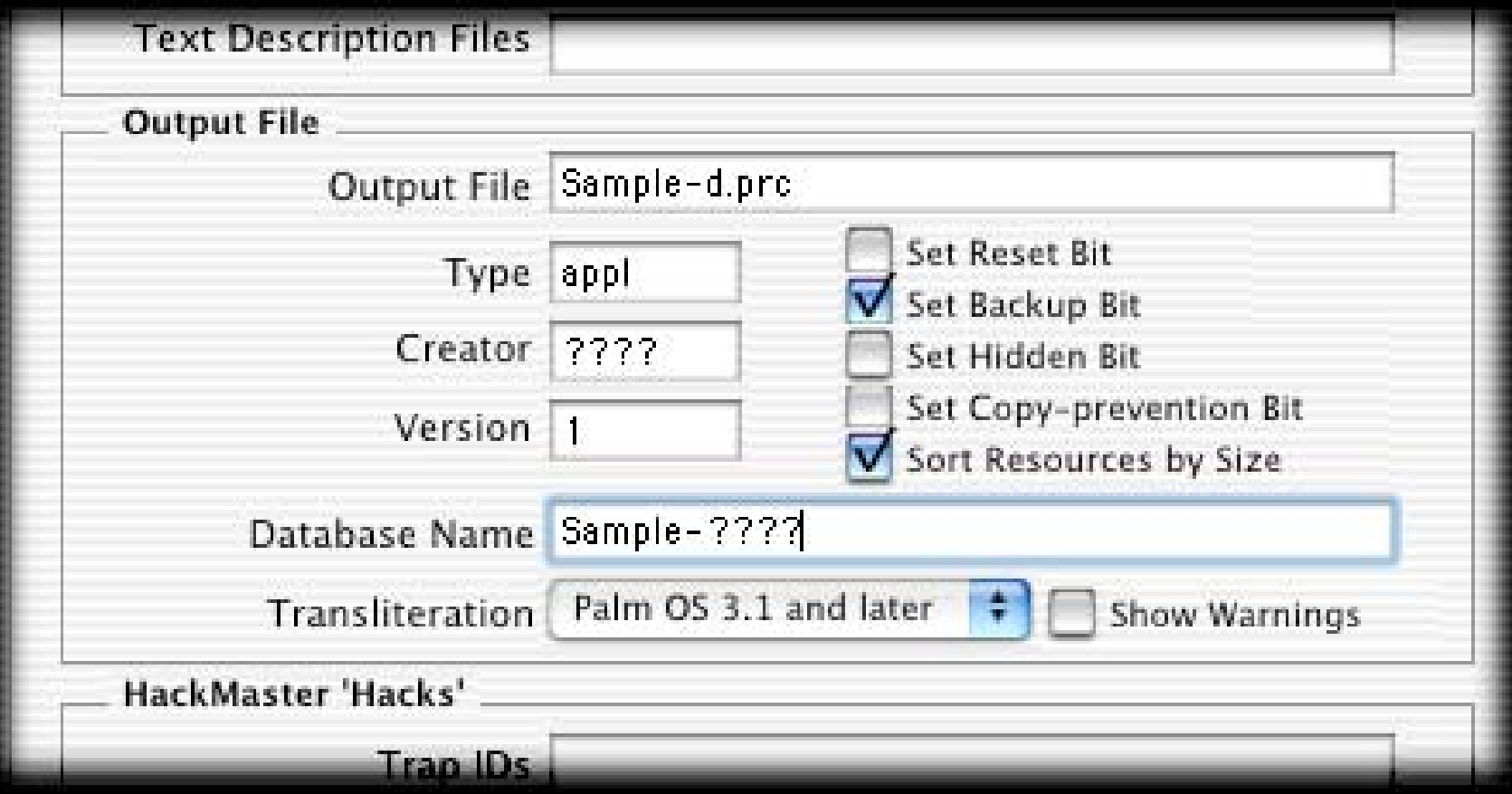
- Code Model
  - Small
  - Smart
- MacsBug Symbols
  - Debug build: on (new style)
  - Release build: off
- Generate A6 stack frames
  - Debug build: on
  - Release build: off



Target Setting—Global Optimizations

- Turn off for debug builds
  - What you see may not be what you get
  - May prevent setting of breakpoint at statement
- Test after turning on for release builds!
  - Optimizations can have unintended side effects

The PalmRez Post-Linker





The Editor: More Than Meets the Eye

- Symbol completion and function lookup
  - Find Definition
  - Find Reference
- Key bindings
  - Explore
  - Customize
- Split screen
- Finding differences
  - Compare OS sources



Editing Tips

- Bring source up to date before adding code to a new file
  - Enables Header popup menu
  - Browser colors
- Functions popup
  - Control-click to sort by function name
  - Use `#pragma mark` to organize
- Disassemble to look for really difficult problems



Searching

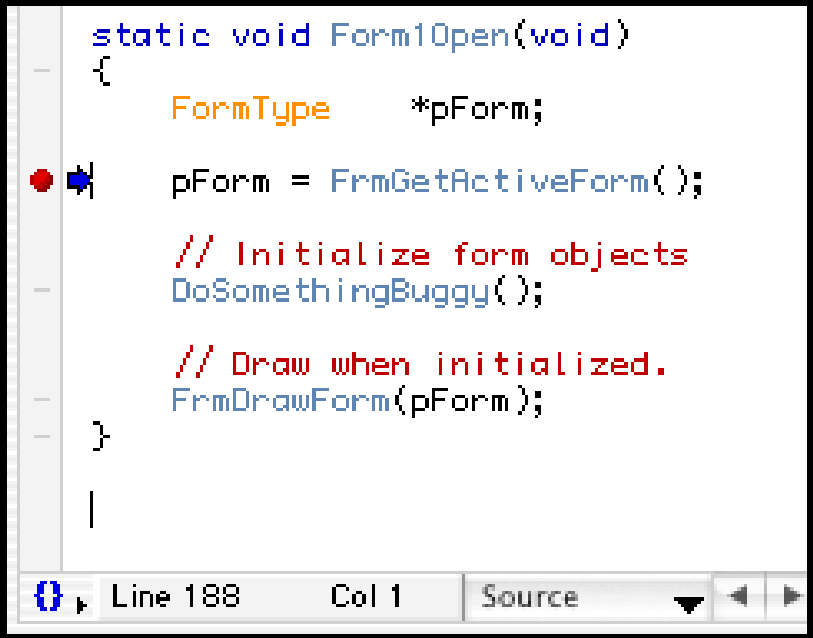
- Search file sets or folders
  - Provides context for API functions
  - “I know I’ve done this before…”
- Candidate sets:
  - Palm OS® Version 4.0 Examples
  - Palm OS Version 4.0 Headers
  - Palm OS Source Code
  - Project Directories

Debugging Tips

- Tracing/logging
  - `HostControl.h`
  - `DbgMgr.h`
  - `DlkServer.h`
- Expressions Window
  - View variables in different formats simultaneously
  - View locals from calling functions in the stack
- Console
  - Fill up memory
  - Examine heaps

Debugging Tips—Source Pane

- Drag the current expression arrow to skip over statements
- Alt-click to set temporary breakpoints
- Breakpoints missing? Turn off optimizations
- Mixed source and assembly





Debugging Tips— Breakpoints Window

- See all breakpoints in one window
- Disable breakpoints without clearing
- Set conditional breakpoints
- Single watch point
  - But you can set with PalmDebugger **`ss`** command



Disassemble Your Source

- Can provide many answers unavailable from other sources
  - Use of global variables (look for A5)
  - Where are your constants stored?
  - Why does my code fail with optimization turned on?
- Takes time to learn
  - Don’t need to learn all the opcodes
  - Get Motorola 68K manual
  - Attend Advanced Debugging & Development class

Resources: Adding Private Data

- Use Rez or PilRC
- Example: include .jpg file as resource

```
type 'tJPG' {
  hex string;
};
READ 'tJPG' (1000)  "MyGraphic.jpg";
```

- Save file with .r suffix

```
hJpg = DmGet1Resource('tJPG', 1000);
pJpg = MemPtrLock(hJpg);
```

Turn Constructor File to Rez File

- Source control
- Cross-platform compilation
- Steps
  1. Add UIResDefs.r as Rez panel prefix file
  2. Change File Mappings of resource files to use Rez as compiler
  3. Right-click on resource file and choose Disassemble command



Constructor Tips

- Get the latest version
  - OS 4.0 SDK Update 1
- Build resource templates
  - Common alerts and forms
  - Difficult-to-create resources (repeating buttons)
- Avoid invalid resources
  - Icon families and older icon types

Documentation Tips

- Palm Documentation
  - Now integrated into IDE!
  - Get latest docs from www.palmos.com
  - Palm OS Companion
  - Development Tools Guide
  - Palm OS Reference
- CodeWarrior Documentation
  - Targeting Palm OS®
  - IDE User Guide



Summary

- Read the documentation
- Browse the Knowledge Base
- www.palmoswerks.com
- Developer courses
  - Advanced Development and Debugging
  - Palm OS Communications



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